CANON: An AI-Assisted Reconciliation Engine for CRM-Data Warehouse Hierarchy Synchronization

Abstract—Organizations that run a customer relationship management (CRM) system such as Salesforce or HubSpot alongside a downstream data warehouse such as Databricks or Snowflake maintain two independent records of account ownership and reporting hierarchy. These records drift out of sync as territories are reassigned, employees leave, and accounts are re-parented, producing incorrect commission payouts, unreliable forecasts, and disputed account ownership. Existing data-quality and ETL tooling can detect such mismatches but does not resolve them, leaving resolution to manual, cross-system comparison. This paper presents a reconciliation pipeline that connects to a CRM and a warehouse source, normalizes both into a canonical entity schema, and performs embedding-based blocking to generate candidate cross-source matches without exhaustive pairwise comparison. Field level conflicts between matched entities are surfaced with a rule-based severity score, and a large language model agent, orchestrated as a state machine and prompted with configurable survivorship rules, proposes a resolution together with a natural-language rationale and a confidence score. Low-confidence or rule-ambiguous cases are routed to a human reviewer, who may approve, escalate, or override each proposal; every decision, automated or human, is appended to an immutable audit log. The pipeline is evaluated on a synthetically generated organizational hierarchy with seeded, labeled conflicts against two baselines, a rule-only matcher and a fuzzy-string matcher, using precision, recall, and F1 over both conflict detection and resolution correctness. Pipeline stages are pure functions, so the evaluation harness exercises the same code paths as production, keeping reported results reproducible across runs.

Index Terms—entity resolution, embedding-based blocking, conflict detection, large language model agents, human-in-the-loop review, master data reconciliation, audit trails

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